

Pedestrians Movement on Crossings

Proof of Concept

Prototype demonstration with screenshots

Course: Agent Systems

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1. Prototype overview

This document presents a working Proof of Concept (PoC) for the pedestrian movement simulation based on the Social Force Model. The prototype implements the core agent behaviors described in the Agent Definition document and demonstrates two key scenarios: a bottleneck passage and a four-way crossing intersection. The goal is to verify that the fundamental design assumptions are feasible and that the minimal version of the simulation produces realistic crowd dynamics.

2. Architecture

The prototype follows a modular architecture with clear separation of concerns. The class diagram below shows the key components:

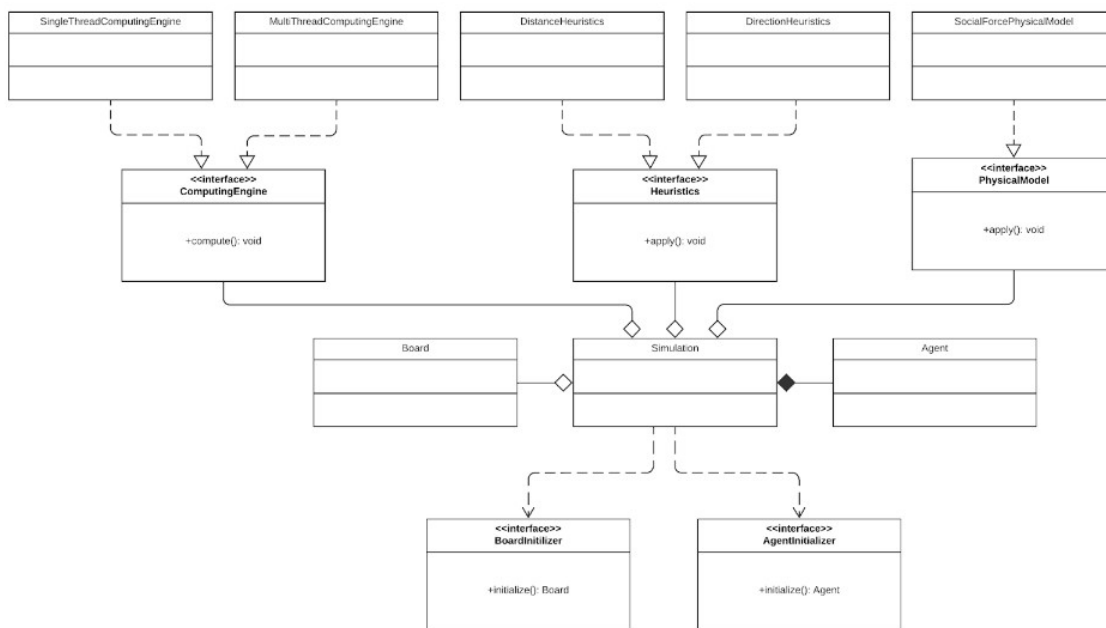


Fig. 1: UML class diagram of the prototype architecture.

The architecture is built around three core interfaces that enable flexible configuration:

- **ComputingEngine** - defines how agent updates are computed. Two implementations are provided: **SingleThreadComputingEngine** (sequential) and **MultiThreadComputingEngine** (parallel using thread pools).
- **Heuristics** - defines behavioral rules applied to agents each step. **DirectionHeuristic** selects the optimal movement direction, while **DistanceHeuristic** adjusts the desired speed based on proximity to obstacles.
- **PhysicalModel** - defines the physics applied after heuristics. **SocialForcePhysicalModel** implements the acceleration equation combining desired velocity, agent-agent collision forces, and agent-wall collision forces.

The **Simulation** class orchestrates the loop: each step applies heuristics to all agents, then applies the physical model, and finally advances agent state using a double-buffer mechanism.

3. Configuration

Before starting a simulation, the user configures its parameters through a dedicated configuration window:

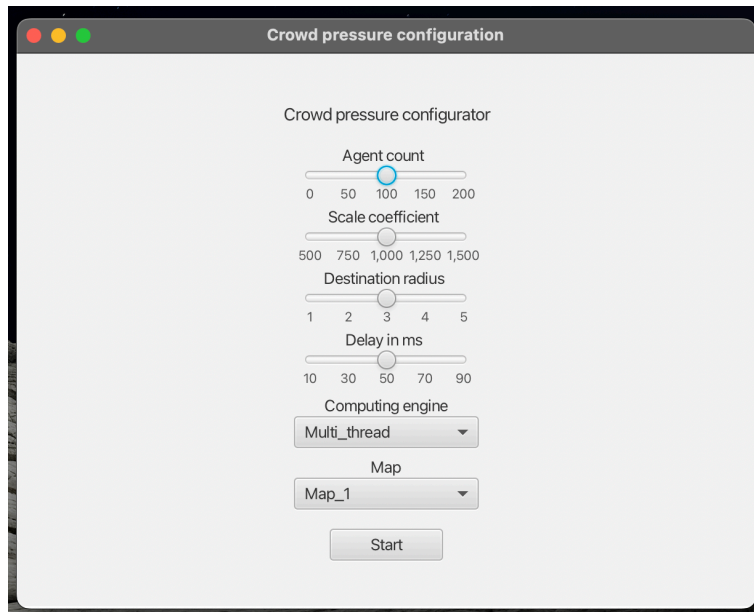


Fig. 2: Configuration window.

The following parameters are available:

- Agent count - the number of pedestrian agents (0-200).
- Scale coefficient - controls the magnitude of physical collision forces.
- Destination radius - the radius around the goal within which an agent is considered to have arrived.
- Delay in ms - the time interval between simulation steps (controls animation speed).
- Computing engine - single-threaded or multi-threaded execution.
- Map - selects the scenario layout (wall configuration and agent spawn positions).

4. Scenario: Bottleneck passage

The first scenario demonstrates a group of 110 agents moving through a narrow gap in a wall. This validates the core Social Force Model mechanics: goal-directed movement, obstacle avoidance, and physical collision handling.

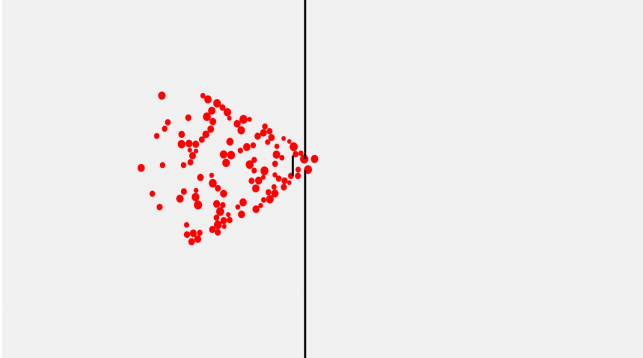


Fig. 3a: Agents approach the wall.

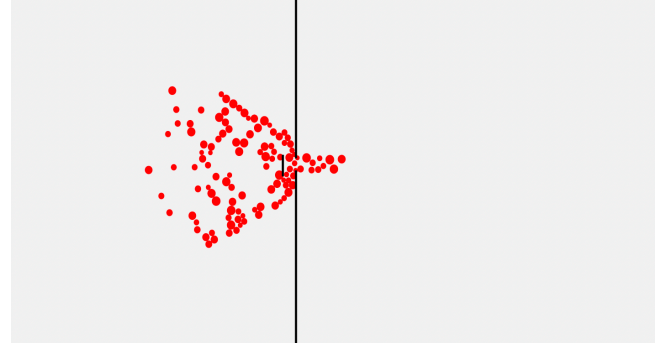


Fig. 3b: Crowd accumulates at the gap.

Agents begin on the left side and move towards a destination on the right. As they approach the wall, the direction heuristic steers them towards the narrow opening. Agents accumulate in front of the gap, forming a characteristic arching cluster - a well-known bottleneck congestion pattern.

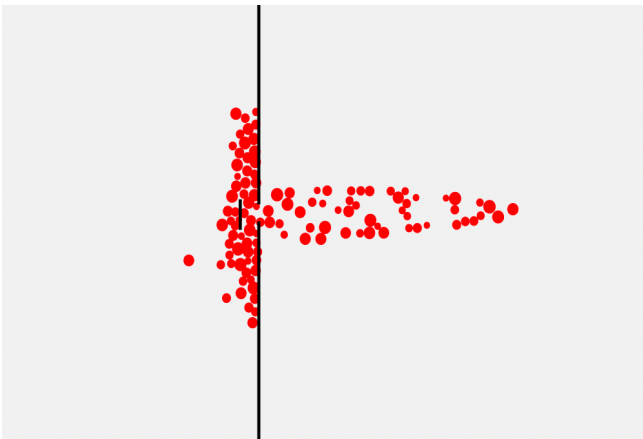


Fig. 3c: Agents pass through the gap.

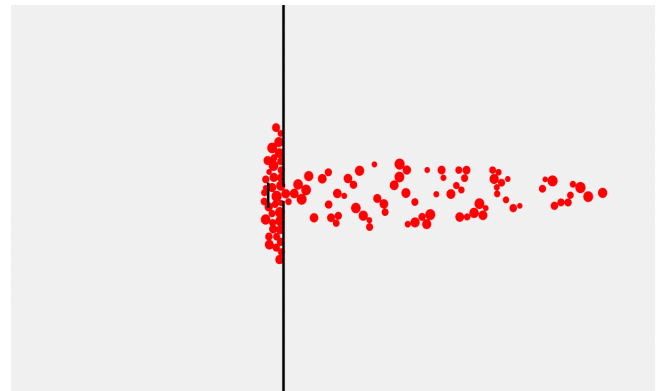


Fig. 3d: Most agents have crossed.

Agents gradually pass through the bottleneck one by one or in small groups. The collision forces prevent agents from overlapping, while the distance heuristic causes agents to slow down as they approach the congested area. After crossing, agents spread out and continue towards the destination. This scenario successfully demonstrates the "freezing by heating" effect - when too many agents push simultaneously, the flow through the gap temporarily stalls.

5. Next steps

Based on the PoC results, the following steps are planned:

- Implement multi-group scenarios - introduce color-coded agent groups with different origins and destinations, as outlined in the Agent Definition document (e.g., four groups approaching a crossing from cardinal directions).
- Add more map layouts - extend beyond the single bottleneck scenario to include multi-way crossing intersections and corridor configurations.
- Implement crowd pressure visualization - color-code agents based on the physical forces they experience, making

it possible to visually identify high-pressure zones as described in the project scope.

- Enable runtime parameter switching - allow toggling heuristics and switching between single-threaded and multi-threaded computation during a running simulation.
- Validate emergent phenomena - verify that the expected behaviors identified in the State of the Art review (lane formation, freezing by heating, group mixing/separation) emerge in the implemented scenarios.